COLLEGE OF ENGINEERING & TECHNOLOGY

Department: Basic and Applied Sciences

EBA0201 – PRECALCULUS

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

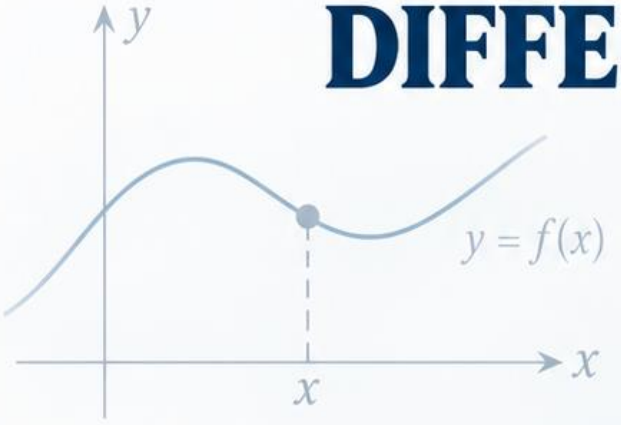
Lecture 2

BASIC RULES OF DIFFERENTIATION

$$\frac{d}{dx}[x^n] = nx^{n-1}$$

$$\frac{d}{dx}[\sin x] = \cos x$$

$$\frac{d}{dx}[\cos x] = -\sin x$$





Build
Stronger Foundations
for a Brighter Tomorrow

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Derivative notation

If $y = f(x)$, the derivative of y with respect to x is written as:

$$\boxed{y' = \frac{dy}{dx}}$$

Constant and power rules

$$\boxed{\frac{d}{dx}(c) = 0}$$

$$\boxed{\frac{d}{dx}(x^n) = nx^{n-1}}$$

Sum and difference

$$\boxed{\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)}$$

Product rule

$$\boxed{y = uv \quad \Rightarrow \quad y' = u'v + uv'}$$

Quotient rule

$$\boxed{y = \frac{u}{v} \quad \Rightarrow \quad y' = \frac{vu' - uv'}{v^2}}$$

Chain rule

Use the chain rule when one function is inside another.

$$\boxed{[f(u)]' = f'(u)u'}$$

Example 1

Find $\frac{dy}{dx}$: $y = 7$

Solution:

$$\boxed{y' = 0}$$

Example 2

Find $\frac{dy}{dx}$: $y = 25$

Solution:

$$\frac{dy}{dx} = 0$$

Example 3

Find $\frac{dy}{dx}$: $y = -15$

Solution:

$$\frac{dy}{dx} = 0$$

Example 4

Find $\frac{dy}{dx}$: $y = 100$

Solution:

$$\frac{dy}{dx} = 0$$

Example 5

Find $\frac{dy}{dx}$: $y = \pi$

Solution:

$$\frac{dy}{dx} = 0$$

Example 6

Find $\frac{dy}{dx}$:

$$y = 6x^5$$

Solution:

$$\frac{dy}{dx} = 6\frac{d}{dx}\left(x^5\right) = 6(5x^4) = 30x^4$$

Example 7

Find $\frac{dy}{dx}$: $y = -4x^7$

Solution:

$$\frac{dy}{dx} = -4\frac{d}{dx}(x^7) = -4(7x^6) \boxed{= -28x^6}$$

Example 8

Find $\frac{dy}{dx}$: $y = x^{-3}$

Solution:

$$y' = -3x^{-4} = \boxed{-\frac{3}{x^4}}$$

Example 9

Find $\frac{dy}{dx}$: $y = \sqrt{x} = x^{1/2}$

Solution:

$$y' = \frac{1}{2}x^{-1/2} = \boxed{\frac{1}{2\sqrt{x}}}$$

Example 10

Find $\frac{dy}{dx}$: $y = 3x^4 - 5x^2 + 7x - 9$

Solution:

$$\boxed{y' = 12x^3 - 10x + 7}$$

Example 11

Find $\frac{dy}{dx}$: $y = 7x^4 + 3x^3 - 5x^2 + 8x - 6$

Solution:

$$\frac{dy}{dx} = \frac{d}{dx}(7x^4) + \frac{d}{dx}(3x^3) - \frac{d}{dx}(5x^2) + \frac{d}{dx}(8x) - \frac{d}{dx}(6)$$
$$\boxed{= 28x^3 + 9x^2 - 10x + 8}$$

Example 12

Find $\frac{dy}{dx}$:

$$y = 4x^6 - 2x^5 + 9x^3 - 7x + 10$$

Solution:

$$\frac{dy}{dx} = 4(6x^5) - 2(5x^4) + 9(3x^2) - 7(1) + 0 \boxed{= 24x^5 - 10x^4 + 27x^2 - 7}$$

Example 13

Find $\frac{dy}{dx}$:

$$y = 9x^9 - 4x^5 + 6x^3 - 12x + 20$$

Solution:

$$\frac{dy}{dx} = 9(9x^8) - 4(5x^4) + 6(3x^2) - 12$$

Example 14

Find $\frac{dy}{dx}$: $y = 15x^7 - 8x^4 + 2x^2 - 5$

Solution:

$$\frac{dy}{dx} = 15(7x^6) - 8(4x^3) + 2(2x) \boxed{= 105x^6 - 32x^3 + 4x}$$

Example 15

Find $\frac{dy}{dx}$: $y = 12x^5 + 5x^4 - 3x^2 + 11$

Solution:

$$\frac{dy}{dx} = 12(5x^4) + 5(4x^3) - 3(2x) + 0 \boxed{= 60x^4 + 20x^3 - 6x}$$

Example 16

Find $\frac{dy}{dx}$: $y = 3x^8 - 6x^6 + 4x^3 - 9x$

Solution:

$$\frac{dy}{dx} = 3(8x^7) - 6(6x^5) + 4(3x^2) - 9(1) \boxed{= 24x^7 - 36x^5 + 12x^2 - 9}$$

Example 17

Find $\frac{dy}{dx}$: $y = 8(3x^4 - 2x^2 + 5)$

Solution:

$$\frac{dy}{dx} = 8(12x^3 - 4x) \boxed{= 96x^3 - 32x}$$

Example 18

Find $\frac{dy}{dx}$: $y = -5(2x^6 + x^3 - 4x)$

Solution:

$$\frac{dy}{dx} = -5(12x^5 + 3x^2 - 4) \boxed{= -60x^5 - 15x^2 + 20}$$

Example 19

Find $\frac{dy}{dx}$: $y = 10(x^5 - 3x^2 + 7)$

Solution:

$$\frac{dy}{dx} = 10(5x^4 - 6x) \boxed{= 50x^4 - 60x}$$

Example 20

Find $\frac{dy}{dx}$: $y = 3(4x^7 - 5x^3 + x^2 - 1)$

Solution:

$$\frac{dy}{dx} = 3(28x^6 - 15x^2 + 2x) \boxed{= 84x^6 - 45x^2 + 6x}$$

Example 21

Find $\frac{dy}{dx}$: $y = (3x^2 + 1)^5$

Solution:

$$y' = 5(3x^2 + 1)^4(6x) = \boxed{30x(3x^2 + 1)^4}$$

Example 22

Find $\frac{dy}{dx}$: $y = (x^2 + 1)^3$

Solution:

$$\frac{dy}{dx} = 3(x^2 + 1)^2(2x) \boxed{= 6x(x^2 + 1)^2}$$

Example 23

Find $\frac{dy}{dx}$: $y = (2x^3 - 5)^2$

Solution:

$$\frac{dy}{dx} = 2(2x^3 - 5)(6x^2) \boxed{= 12x^2(2x^3 - 5)}$$

Example 24

Find $\frac{dy}{dx}$: $y = (x^4 + x)^4$

Solution:

$$\frac{dy}{dx} = 4(x^4 + x)^3(4x^3 + 1) \boxed{= 4(x^4 + x)^3(4x^3 + 1)}$$

Example 25

Find $\frac{dy}{dx}$: $y = (3x^2 - 2x + 1)^5$

Solution:

$$\frac{dy}{dx} = 5(3x^2 - 2x + 1)^4(6x - 2) \boxed{= 5(3x^2 - 2x + 1)^4(6x - 2)}$$

Example 26

Find $\frac{dy}{dx}$: $y = (x^3 - 4x^2 + 2)^3$

Solution:

$$\frac{dy}{dx} = 3(x^3 - 4x^2 + 2)^2(3x^2 - 8x) \boxed{= 3(x^3 - 4x^2 + 2)^2(3x^2 - 8x)}$$

Example 27

Find $\frac{dy}{dx}$: $y = x^2(x^3 + 1)$

Solution:

$$y' = 2x(x^3 + 1) + x^2(3x^2) \boxed{= 5x^4 + 2x}$$

Example 28

Find $\frac{dy}{dx}$: $y = (x^2 + 3)(x^3 + 2)$

Solution:

$$\frac{dy}{dx} = (x^2 + 3)(3x^2) + (x^3 + 2)(2x) = 3x^4 + 9x^2 + 2x^4 + 4x \boxed{= 5x^4 + 9x^2 + 4x}$$

Example 29

Find $\frac{dy}{dx}$: $y = (2x^3 + x)(x^2 - 5)$

Solution:

$$\frac{dy}{dx} = (2x^3 + x)(2x) + (x^2 - 5)(6x^2 + 1) = 4x^4 + 2x^2 + 6x^4 - 29x^2 - 5 \boxed{= 10x^4 - 27x^2 - 5}$$

Example 30

Find $\frac{dy}{dx}$: $y = (5x^3 - x^2)(x^4 + 2)$

Solution:

$$\frac{dy}{dx} = (5x^3 - x^2)(4x^3) + (x^4 + 2)(15x^2 - 2x)$$

$$= 20x^6 - 4x^5 + 15x^6 - 2x^5 + 30x^2 - 4x \boxed{= 35x^6 - 6x^5 + 30x^2 - 4x}$$

Example 31

Find $\frac{dy}{dx}$: $y = (x^5 + 3x)(2x^3 - x)$

Solution:

$$\frac{dy}{dx} = (x^5 + 3x)(6x^2 - 1) + (2x^3 - x)(5x^4 + 3)$$
$$= 6x^7 + 18x^3 - x^5 - 3x + 10x^7 - 5x^5 + 6x^3$$
$$- 3x \boxed{= 16x^7 - 6x^5 + 24x^3 - 6x}$$

Example 32

Find $\frac{dy}{dx}$: $y = (4x^2 - 7)(x^3 + x)$

Solution:

$$\frac{dy}{dx} = (4x^2 - 7)(3x^2 + 1) + (x^3 + x)(8x)$$
$$= 12x^4 - 17x^2 - 7 + 8x^4 + 8x^2 \boxed{= 20x^4 - 9x^2 - 7}$$

Example 33

Find $\frac{dy}{dx}$: $y = (x^4 - 2x)(3x^2 + 1)$

Solution:

$$\frac{dy}{dx} = (x^4 - 2x)(6x) + (3x^2 + 1)(4x^3 - 2)$$
$$= 6x^5 - 12x^2 + 12x^5 + 4x^3 - 6x^2 - 2 \boxed{= 18x^5 + 4x^3 - 18x^2 - 2}$$

Example 34

Find $\frac{dy}{dx}$:

$$y = \frac{x^2 + 1}{x - 1}$$

Solution:

$$y' = \frac{(x - 1)(2x) - (x^2 + 1)}{(x - 1)^2} \boxed{= \frac{x^2 - 2x - 1}{(x - 1)^2}}$$

Example 35

Find $\frac{dy}{dx}$:

$$y = \frac{x^2 + 2}{x + 1}$$

Solution:

$$\frac{dy}{dx} = \frac{(x + 1)(2x) - (x^2 + 2)(1)}{(x+1)^2} \boxed{= \frac{x^2 + 2x - 2}{(x+1)^2}}$$

Example 36

Find $\frac{dy}{dx}$:

$$y = \frac{3x^3 - x}{x^2 + 4}$$

Solution:

$$\frac{dy}{dx} = \frac{(x^2 + 4)(9x^2 - 1) - (3x^3 - x)(2x)}{(x^2+4)^2} \boxed{= \frac{3x^4 + 38x^2 - 4}{(x^2+4)^2}}$$

Example 37

Find $\frac{dy}{dx}$:

$$y = \frac{x^4 + 5x^2}{x^3 - 2}$$

Solution:

$$\frac{dy}{dx} = \frac{(x^3 - 2)(4x^3 + 10x) - (x^4 + 5x^2)(3x^2)}{(x^3-2)^2} \boxed{= \frac{x^6 - 10x^3 - 20x}{(x^3-2)^2}}$$

Example 38

Find $\frac{dy}{dx}$:

$$y = \frac{2x^5 - x}{3x^2 + 1}$$

Solution:

$$\frac{dy}{dx} = \frac{(3x^2 + 1)(10x^4 - 1) - (2x^5 - x)(6x)}{(3x^2+1)^2} \boxed{= \frac{18x^6 + 10x^4 + 1}{(3x^2+1)^2}}$$

Example 39

Find $\frac{dy}{dx}$:

$$y = \frac{x^6 - 4x^3}{x^2 + 5}$$

Solution:

$$\frac{dy}{dx} = \frac{(x^2 + 5)(6x^5 - 12x^2) - (x^6 - 4x^3)(2x)}{(x^2+5)^2} \boxed{= \frac{4x^7 + 30x^5 - 60x^2}{(x^2+5)^2}}$$

Example 40

Find $\frac{dy}{dx}$:

$$y = \frac{5x^4 + 2x}{x^3 - x + 1}$$

Solution:

$$\frac{dy}{dx}$$

$$= \frac{(x^3 - x + 1)(20x^3 + 2) - (5x^4 + 2x)(3x^2 - 1)}{(x^3 - x + 1)^2} \boxed{= \frac{5x^6 - 15x^4 + 20x^3 - 4x^2 + 2}{(x^3 - x + 1)^2}}$$